

LDA IMPLEMENTATION:

Implement **Linear Discriminant Analysis (LDA)** from scratch in **Python** without using any built-in LDA functions (e.g., `sklearn.discriminant_analysis.LinearDiscriminantAnalysis`).

Your implementation should:

1. Compute the class-wise means and the overall mean.
2. Construct the Within-Class Scatter Matrix (S_W).
3. Construct the Between-Class Scatter Matrix (S_B).
4. Solve the generalized eigenvalue problem for $S_W^{-1}S_B$.
5. Select the top k discriminant vectors based on the largest eigenvalues.
6. Project the given dataset onto the new lower-dimensional subspace.
7. Visualize the transformed data for a two-class or multi-class dataset.

Restrictions:

- You may use NumPy for matrix operations and Matplotlib for visualization.
- Do not use any built-in LDA implementation from Scikit-learn or other machine learning libraries.
- Clearly comment each step of your code and explain the role of S_W , S_B , eigenvalues, and eigenvectors in the LDA process.

Expected Output:

- Reduced-dimensional feature matrix.
- Plot showing the projected samples.
- Brief explanation of how LDA maximizes class separability.